Moving Oracle Integration Connections to Fusion onto OAuth JWT User Assertion, Including BI Publisher

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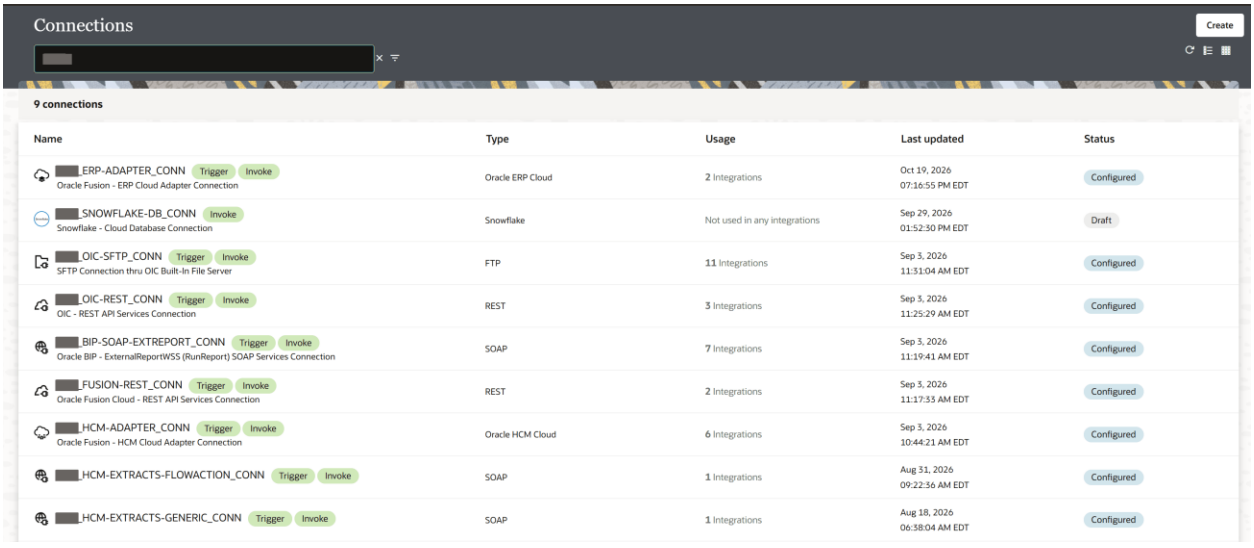
Product: Oracle Integration 3 (REST, SOAP, Oracle ERP Cloud and Oracle HCM Cloud adapters), Oracle Fusion Cloud Applications, BI Publisher

Note: environment identifiers (OCIDs, client IDs, identity domain and pod identifiers, and email addresses) are masked in the screenshots.

Oracle is retiring password-based authentication for Fusion REST and SOAP services, with migration expected before March 31, 2027. With the signing key and identity domain trust in place, I switched every Oracle Integration connection that calls Fusion to OAuth using JWT User Assertion. Two findings stood out: BI Publisher accepted JWT User Assertion, and the ERP and HCM Cloud adapters must be saved before they can be tested.

Part 1: Inventory the connections

I filtered the Connections page to list every connection that calls Fusion. Six were in scope: a REST connection, a BI Publisher SOAP connection, two SOAP connections for HCM extract services, an Oracle ERP Cloud adapter connection and an Oracle HCM Cloud adapter connection. I switched them one at a time, recording each one's previous policy so I could roll back.

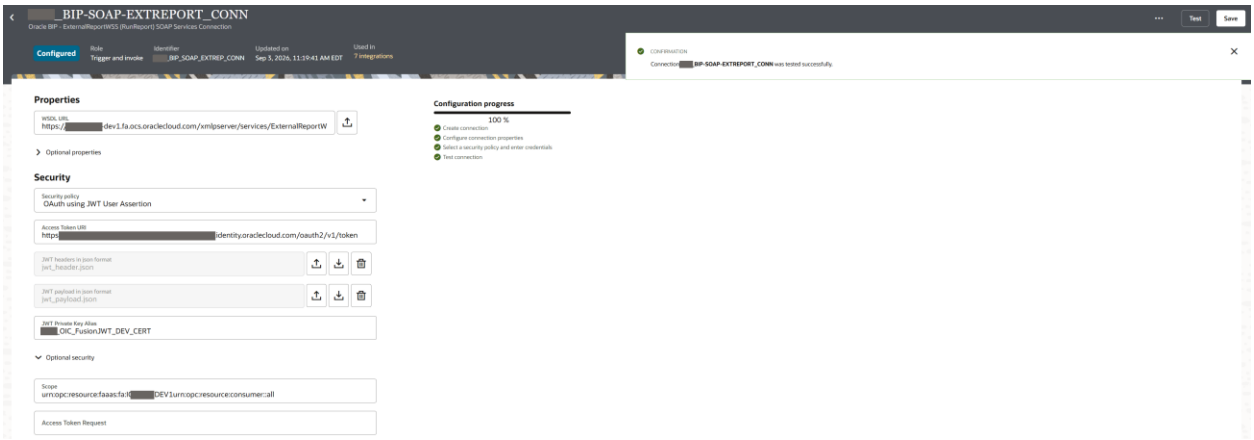


The connections that call Fusion, with how many integrations use each.

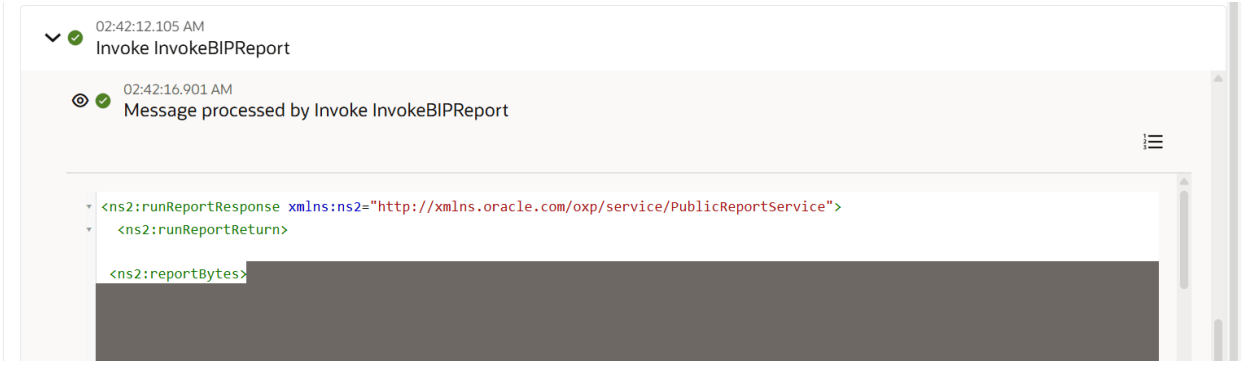
Part 2: Switch the BI Publisher SOAP connection and prove it with a run

I changed the security policy to OAuth using JWT User Assertion and entered the identity domain's token URI, the JWT header file (alg RS256, typ JWT, kid set to the key alias), the JWT payload file (iss set to the client ID, sub set to the integration user, aud https://identity.oraclecloud.com/), the private key alias and the scope. I left Access Token Request blank. Test succeeded.

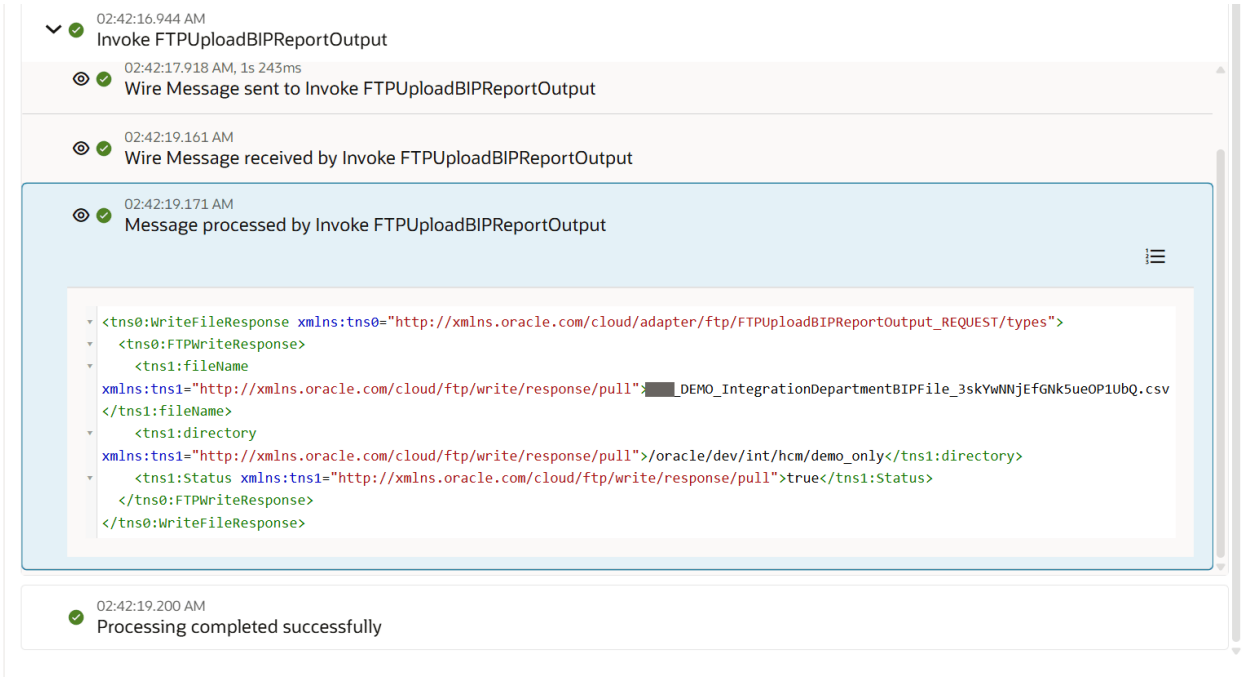
A successful Test on a SOAP connection only proves the WSDL and the token, so I ran an integration that calls ExternalReportWSSService runReport. The response returned report bytes, and the integration wrote a 10 KB CSV to the Oracle Integration file server. This showed that BI Publisher works with JWT User Assertion, so an Authorization Code application is not needed for it.



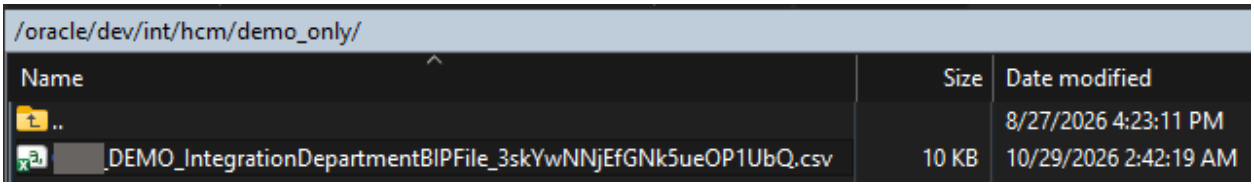
The BI Publisher SOAP connection on OAuth using JWT User Assertion, tested successfully.



runReport response from BI Publisher with report bytes (content masked).



The report output written to the Oracle Integration file server; processing completed successfully.



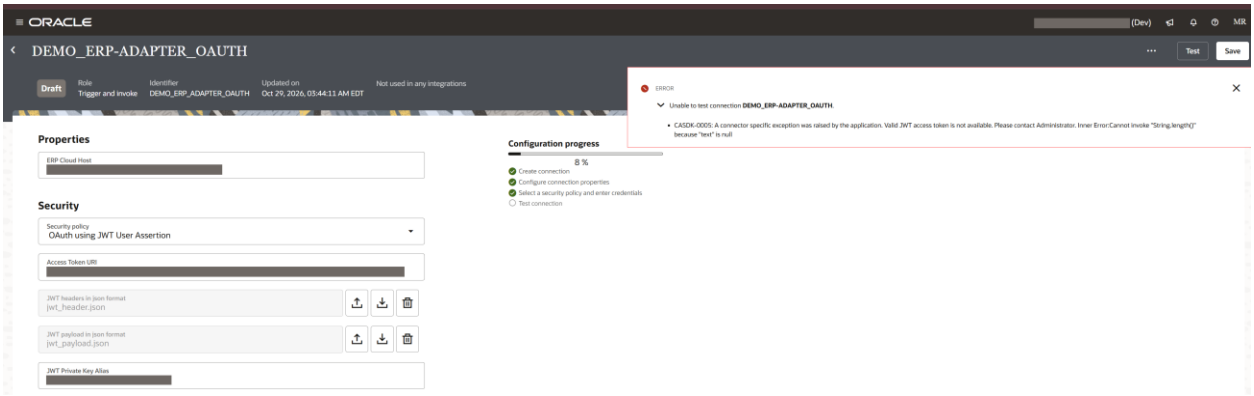
The 10 KB output file in the file server folder.

Part 3: The ERP and HCM Cloud adapters need Save before Test

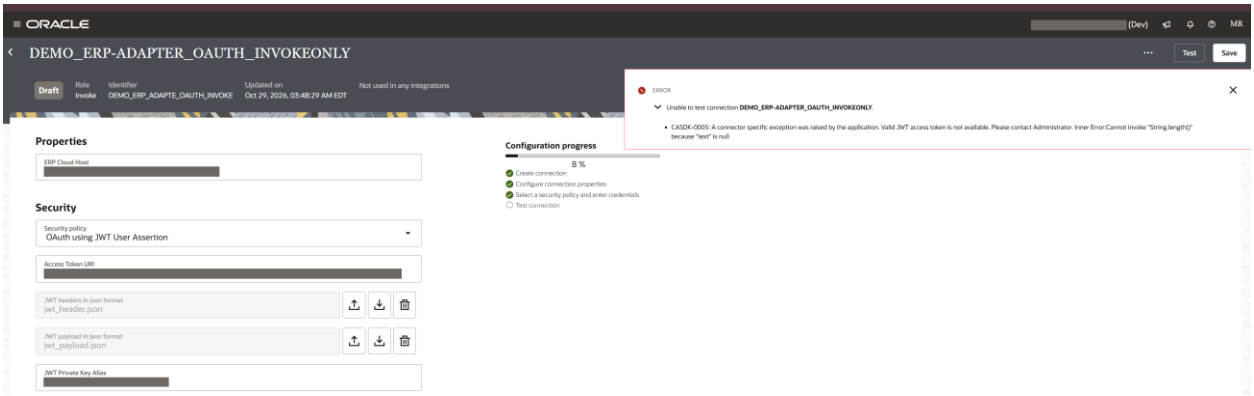
The REST and two HCM extract SOAP connections tested successfully the same way. The ERP Cloud and HCM Cloud adapter connections did not. With the same values, Test failed with CASDK-0005, "Valid JWT access token is not available", with an inner error that a text value was null.

I re-uploaded the JWT files, reset the policy, and created a new ERP Cloud adapter connection, first with the Trigger and Invoke role and then Invoke only. All failed the same way. The new connection then tested successfully after it had been saved. These two adapters test the saved configuration rather than the values on screen. The order that works is: enter the values, Save (status Draft), Test, then Save again (status Configured). REST and SOAP connections can be tested before saving.

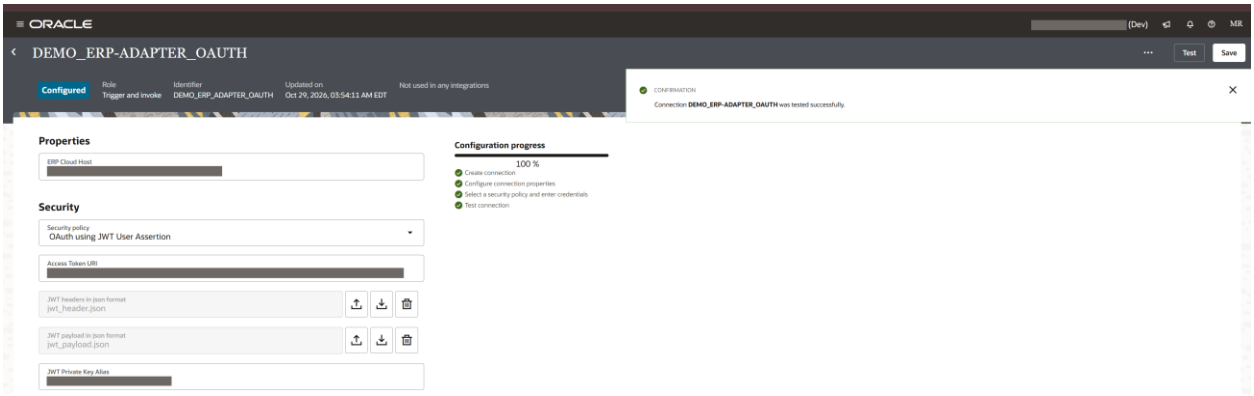
Because the first Save switches a live connection, I did it in a quiet window with the old credentials ready to roll back.



A new ERP Cloud adapter connection failing Test with CASDK-0005 before it was saved.



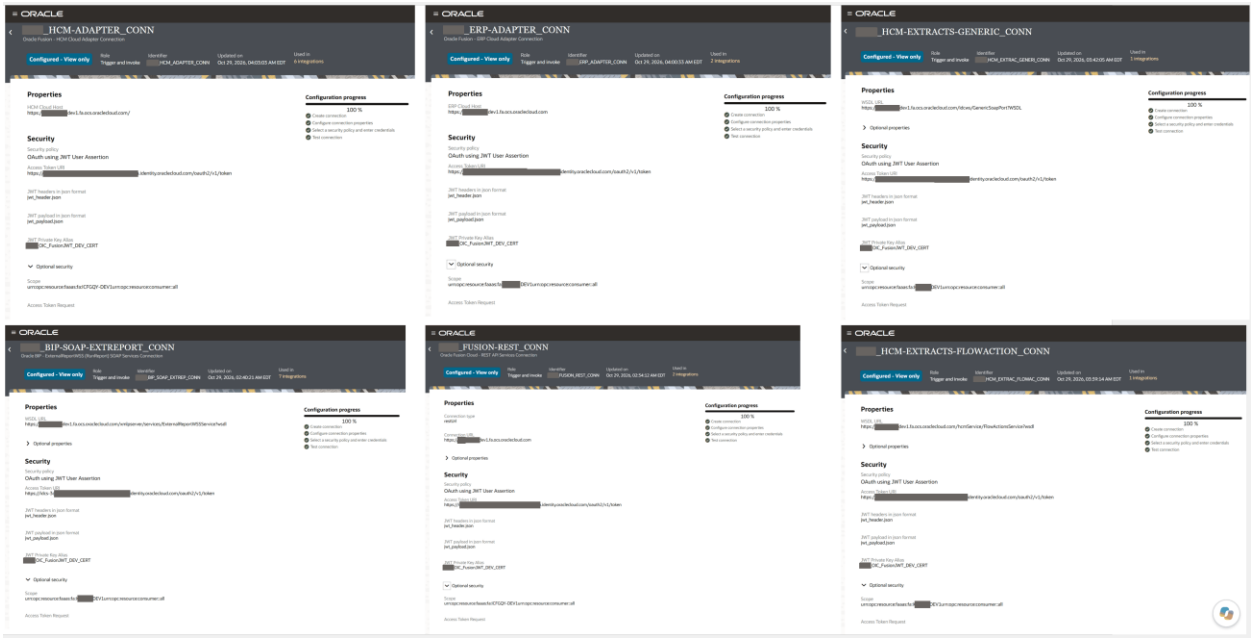
The same error with an Invoke-only connection.



After saving, the same connection tests successfully and shows Configured.

Part 4: All connections on JWT User Assertion

All six connections now show Configured with OAuth using JWT User Assertion, the same token URI, JWT files, key alias and scope. The remaining checks are a successful run on each connection and, for integrations that subscribe to ERP business events or HCM Atom feeds, reactivating them and confirming the next event or poll.



All six connections to Fusion configured with OAuth using JWT User Assertion.

What I learned

One signing key, one confidential application and one set of JWT files covered every type of Fusion connection, including BI Publisher. A Test result is not the same as a working integration, so I proved BI Publisher with a real run. The Save-before-Test behaviour of the ERP and HCM Cloud adapters looked like a product defect at first; testing a fresh connection after saving is what revealed the cause.